

Lecture 14

Blade dynamic response

Part 2

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This lecture

- Modal participation ODEs with harmonic excitation
- Multi-harmonic representation of periodic blade loads
- The steady-state flapping response synthesis
- Blade response control through design
- Internal blade loads

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The flapping response due to $k\Omega$ -excitation

Before considering a more general scenario, the steady-state solution is obtained for the previously established ODE governing the j -th modal participation factor under the harmonic excitation at $k\Omega$:

$$\ddot{q}_j + q_j \omega_{F,j}^2 = \left(L_{j,k} / M_j \right) e^{i(k\Omega)t} \quad q_j = Q_j e^{i(k\Omega)t}, \quad \ddot{q}_j = -(k\Omega)^2 Q_j e^{i(k\Omega)t}$$

Assumed harmonic response

$$(\omega_{F,j}^2 - (k\Omega)^2) Q_j = L_{j,k} / M_j$$

$$Q_j = \frac{1}{\omega_{F,j}^2 - (k\Omega)^2} \frac{L_{j,k}}{M_j}$$

$$q_j(t) = Q_j e^{i(k\Omega)t} = \frac{1}{\omega_{F,j}^2 - (k\Omega)^2} \frac{L_{j,k}}{M_j} e^{i(k\Omega)t}$$

Response of the j -th mode (modal participation) to the excitation at the frequency $k\Omega$ (k -th rotor harmonics).

The overall steady-state flapping response from all participating modes:

$$w(x,t) = \sum_{i=1}^{\infty} q_i(t) w_i(x) = \sum_{i=1}^{\infty} \left(\underbrace{\frac{1}{\omega_{F,i}^2 - (k\Omega)^2}}_{\text{Amplitude of the } j\text{-th modal participation factor}} \underbrace{\frac{\int_{(b)} w_i L_k dx}{M_i}}_{\text{Mode shape function}} \right) w_i(x) e^{i(k\Omega)t}$$

Index j changed to i Mode shape function Oscillation at $k\Omega$

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Overview of the solution strategy

Flapping IV-BVP with periodic L

$$\begin{aligned} (EI w'')'' - (T w')' + m\ddot{w} &= L \\ L(x,t) &= L_k(x) \sin(k\Omega t) \end{aligned}$$

Modal synthesis

$$w(x,t) = \sum_{j=1}^{N_M} q_j(t) w_j(x)$$

BVP

a.k.a. the modal decomposition of the flapping response

The ODE governing the mode participation factor

$$\ddot{q}_j + q_j \omega_{F,j}^2 = \frac{L_{j,k}}{M_j} \sin(k\Omega t) \quad \text{Decoupled ODEs}$$

$$M_j = \int_{(b)} m w_j^2 dx, \quad L_{j,k} = \int_{(b)} w_j L_k dx, \quad \omega_{F,j}^2 = \omega_{0,F,j}^2 + K_{F,j} \Omega^2$$

The steady-state flapping response

$$w(x,t) = \sum_{j=1}^{N_M} \left(\frac{1}{\omega_{F,j}^2 - (k\Omega)^2} \frac{L_{j,k}}{M_j} w_j(x) \right) \sin(k\Omega t)$$

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Periodic and multi-harmonic blade excitation

Previous slides developed the overall steady-state flapping response for the external harmonic loads at the frequency $k\Omega$, i.e., $L(x,t)=L_k(x)\sin(k\Omega t)$. The steady loads (e.g., lift) in rotor systems are periodic in general, e.g., $L(x,t)=L(x,t+2\pi/\Omega)$. The flapping response is therefore sought under assumed general periodic loading expressed as a Fourier series, i.e., a sum of harmonic contributions.

Consider the exponential form of the Fourier series for $L(x,t)$:

$$(EI w'')'' - (T w')' + m\ddot{w} = \sum_{(k)} L_k(x) e^{ik\Omega t}$$

Index k runs from 0 (the constant component) to a chosen maximum value (theoretically infinity, practically around 8 to 12).

Applying the same steps as those introduced for the $k\Omega$ excitation previously, the following ODE governing the k -th mode participation factor is obtained:

$$M_j \ddot{q}_j + (K_j + \Omega^2 K_{CF,j}) q_j = \int_{(b)} w_j \left(\sum_{(k)} L_k(x) e^{ik\Omega t} \right) dx$$

$$\ddot{q}_j + \omega_{F,j}^2 q_j = \frac{1}{M_j} \sum_{(k)} \left(\int_{(b)} w_j L_k dx \right) e^{ik\Omega t}$$

A second order oscillator under multi-harmonic excitation (e.g., $k=0, 1, \dots, 8$). Each mode responds to each frequency!

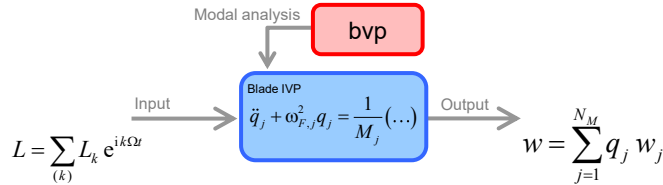
$$w = \sum_{j=1}^{N_M} q_j w_j$$

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The total blade flapping response

The calculation of the steady-state flapping blade response under periodic excitation (IV-BVP) is developed as a process involving solution of the corresponding BVP (modal analysis) and IVP (decoupled harmonic oscillators):



The total flapping response consists of the contributions from **all considered modes** under the action of **all considered harmonic components**. The *principle of superposition* is used to obtain the total flapping response.

A k -th harmonic response component is $w_{\{k\}}(x,t)$. The total response is:

$$w = \sum_{(k)} w_{\{k\}} = \sum_{(k)} \left(\sum_{j=1}^{N_M} \left(\frac{1}{\omega_{F,j}^2 - (k\Omega)^2} \frac{L_{j,k}}{M_j} w_j(x) \right) e^{ik\Omega t} \right)$$

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The total blade flapping response, cont.

The total steady-state response consists of (N_M modal $\times$ N_H harmonic) contributions summed up in the following expression ($k=0$ not considered):

$$w = \sum_{k=1}^{N_H} \sum_{j=1}^{N_M} \left(\frac{1}{\omega_{F,j}^2 - (k\Omega)^2} \frac{L_{j,k}}{M_j} w_j(x) \right) e^{ik\Omega t}$$

This double sum can be visualised using the following table:

	F1	F2	...	FN_M	
1R	$\frac{1}{\omega_{F,1}^2 - (1\Omega)^2} \frac{L_{1,1}}{M_1} w_1 \exp(1\Omega t)$	$\frac{1}{\omega_{F,2}^2 - (1\Omega)^2} \frac{L_{2,1}}{M_2} w_2 \exp(1\Omega t)$	...	$\frac{1}{\omega_{F,N_M}^2 - (1\Omega)^2} \frac{L_{N_M,1}}{M_{N_M}} w_{N_M} \exp(1\Omega t)$	Modal contributions
2R	$\frac{1}{\omega_{F,1}^2 - (2\Omega)^2} \frac{L_{1,2}}{M_1} w_1 \exp(2\Omega t)$	$\frac{1}{\omega_{F,2}^2 - (2\Omega)^2} \frac{L_{2,2}}{M_2} w_2 \exp(2\Omega t)$	...	$\frac{1}{\omega_{F,N_M}^2 - (2\Omega)^2} \frac{L_{N_M,2}}{M_{N_M}} w_{N_M} \exp(2\Omega t)$	
...	...	...	...	...	
$N_H R$	$\frac{1}{\omega_{F,1}^2 - (N_H\Omega)^2} \frac{L_{1,N_H}}{M_1} w_1 \exp(N_H\Omega t)$	$\frac{1}{\omega_{F,2}^2 - (N_H\Omega)^2} \frac{L_{2,N_H}}{M_2} w_2 \exp(N_H\Omega t)$	...	$\frac{1}{\omega_{F,N_M}^2 - (N_H\Omega)^2} \frac{L_{N_M,N_H}}{M_{N_M}} w_{N_M} \exp(N_H\Omega t)$	

Significance of the individual response terms:

$$\frac{1}{\omega_{F,j}^2 - (k\Omega)^2} \frac{L_{j,k}}{M_j} w_j(x) e^{ik\Omega t}$$

Modal amplification! Lift modal forcing coefficient & modal mass Mode shape function Oscillation at $k\Omega$

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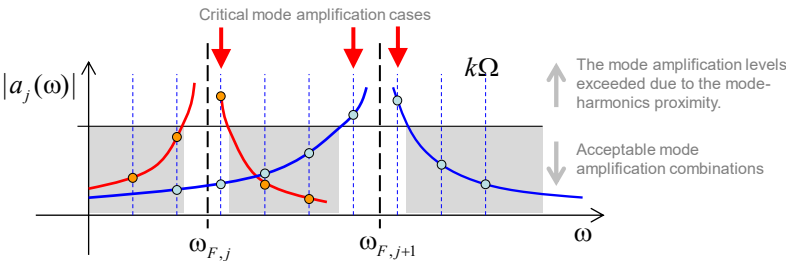
The mode amplification effect

As seen in the pervious slide, the response of a blade can be altered and controlled "by design" when considering the following factors:

- modal amplification parameter $A_{j,k}$
- modal forcing $L_{j,k}$
- modal mass M_j

The mode (or modal) amplification parameter $A_{j,k}$ is visualised with the help of the corresponding mode amplification function $a_j(\omega)$:

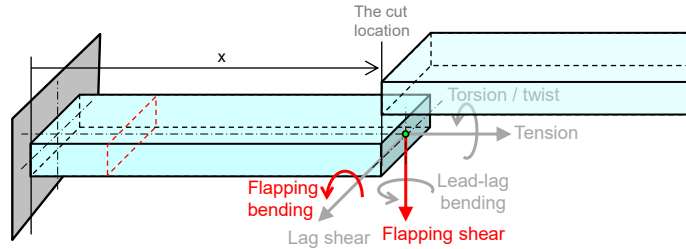
$$a_j(\omega) = \frac{1}{\omega_{F,j}^2 - \omega^2} \Rightarrow A_{j,k} = a_j(k\Omega) = \frac{1}{\omega_{F,j}^2 - (k\Omega)^2}$$



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Internal blade loads

Consider a cut at the blade's radial station x to reveal its internal (sectional) loads:



Knowing the deflection $w(x,t)$, in the next step, we can use the Euler-Bernoulli constitutive relationship to obtain the internal flapping bending moment:

$$M(x,t) = EI_F w''(x,t)$$

Combine $M(x,t)$ with the modal representation of the flapping response:

$$M(x,t) = EI_F w''(x,t) = \sum_{(k)} \sum_{j=1}^{N_M} \left(\frac{EI_F}{\omega_{F,j}^2 - (k\Omega)^2} \frac{L_{j,k}}{M_j} \underbrace{w_j''(x)}_{\text{Modal curvature}} \right) e^{ik\Omega t}$$

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Summary

- Principle of superposition is applied across all considered modes ("modal synthesis").
- Principle of superposition can be applied also across all rotor harmonics ("frequency synthesis").
- The mode amplification is the key dynamic design consideration. Other important design parameters include the modal load, modal mass, mode shapes and their higher order derivatives.

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